INDUSTRIAL IOT & AUTOMATION

CERTIFICATE NUMBER : CERT0610620462

This is to certify that :

Nigel Denaya Manuwu Nayar

Has completed a 32 hours program of INDUSTRIAL IOT & AUTOMATION Training at the IDN.ID Training Center, Jakarta, on 12, 13, 14, 15 Januari 2026
The student was able to obtain proficiency :

Electrical Drawing
a. Mengenal Symbol Standard Electrical Drawing
b. Membaca Electrical Drawing
Wiring
a. Mendesign Wiring diagram
b. Mengaplikasikan Design wiring diagram
c. Wiring dengan symbol, Labeling & Numbering

Pengenalan PLC (Brand, Input, Output, CPU, Interface)
Logika Pemrograman PLC Dasar (Flow Chart, Decision, Alternate Program, & Option)
Mode dalam PLC
a. Write / Program Mode
b. Read Only Mode
c. Edit Online Mode
d. Online Monitoring Mode
Download & Upload Program

Pemrograman PLC lanjutan
a. Compare Instruction
b. Arithmathic Instruction
c. Move Instruction
d. Converstion Word Data
e. Program Analog Input & Output
f. Scalling data Input dan Output (Analog)
g. Communication port (RS232, RS485, Ethernet, CC-Link, Profibus dsb)

Komunikasi PLC dan Instrument
a. Mengenal Berbagai Interface PLC dan HMI
b. Pemrograman Dasar HMI (Bit Lamp, Bit Switch, Bit Word, dsb)
c. Menampilkan Data Bit & Data Word dalam HMII
d. Membuat Program HMII
e. Mengkomunikasikan HMI dan PLCI

Internet of Things
a. Pengenalan IoT
b. Elemen Utama Pendukung IoT
c. Implementasi IoT
d. Pengenalan Mode Data Communication serial
e. Konfigurasi dan Converter ModBus RTU ke ModBus TCP atau Sebaliknya
f. Pengorganisasian data ModBus
g. Frame Data ModBus
h. Konfigurasi Module Modbus RTU
i. Hardware Pendukung IoT Gateway
j. Cloud Platform Connection Configuration
k. Scada Web, atau Display IoT

IoT Project
a. Monitoring & Control Lampu & Penerangan Jalan Umum
b. Monitoring & Control AC
c. Monitoring & Control Fan or Blower
d. Monitoring Temperature & Humidity
e. Monitoring Power Meter (Voltage, Currnet, kWH, Harmonic etc)
f. Monitoring & Control Induction Motor (for Conveyor, Agitator, & Pump)
g. Monitoring & Control Pneumatic System (Valve & Air Pressure)
h. Monitoring & Control Hydraulic System (Valve & Hydro Pressure)
i. Monitoring & Control System Start Delta Motor
j. Monitoring Run & Stop Machine
k. Monitoring Effectivity Machine (Compare of Production Plan & Actual)
l. Monitoring Data Logging pada sebuah Mesin Produksi

15 Januari 2026

Dedi Gunawan
CEO of IDN.ID Training Center